

Virtual Stakeholder System

Group #28

Project Deliverable #3

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1. Introduction:

1.1 Purpose:

The purpose of this SRS document is to define and highlight important requirements specifications for the Virtual Stakeholder System. The Document outlines all requirements, both functional and non-functional, as well as general information and system features. This document's target audience consists of Stakeholders, Product Testers, and Developers.

1.2 Scope of the system:

General features within the scope of the system, and explicit description of what is not in the scope of the system

- a) General Features within the scope of the system:
 - i. The Virtual Stakeholder System will be able to answer questions posed by the user, filling the role of the stakeholder.
 - ii. The Virtual Stakeholder System will allow users to create and manage personas based on LLM technology.
 - iii. The Virtual Stakeholder System will be able to critique requirements submitted by the user as well as provide optimal feedback.
- b) Description of what is not in the scope of the system:
 - i. The Virtual Stakeholder System will not allow two or more collaborators to work together simultaneously in the same stakeholder instance.
 - ii. The Virtual Stakeholder System will not have a voice chat feature.

1.3 Definitions, Acronyms and Abbreviations:

Term	Meaning
US	User Story
VSS	Virtual Stakeholder System
LLM	Large Language Model
SSH	Secure Shell
G	Goal

1.4 Verbal Forms:

1.4.1 - Must

As per IEEE Standard 2018,

The verbal form 'Must' indicates requirements to be followed strictly to conform to this document and from which no deviation is permitted. Must is synonymous with 'shall.'

1.4.2 - Should

As per IEEE Standard 2018,

The verbal form 'should' indicates that a possibility among a set of possibilities is recommended as particularly suitable (without mentioning or excluding other possibilities) or that a certain course of action is preferred but not necessarily required.

1.4.3 - May

As per IEEE Standard 2018,

The verbal form 'may' indicates a course of action permissible within the limits of the document.

1.5 References:

[1] IEEE Standard 29148-2018. (2018). ISO/IEC/IEEE International Standard - Systems and software engineering — Life cycle processes — Requirements engineering. IEEE. <https://doi.org/10.1109/IEEESTD.2018.8559686>

1.6 Goals of the System:

Stakeholder	Goals
1. Requirements Engineer	G1: Efficiently gather and validate requirements from the system G2: Generate accurate project artifacts (personas, user stories, use cases) G3: Ensure requirements coherence and avoid duplicates
2. Project Manager	G4: Track requirement completion status and deadlines G5: Maintain project data security and confidentiality
3. Development Team	G6: Ensure clarity of generated non-functional requirements G7: Interact with virtual stakeholders for requirement clarification G8: Validate requirements against project goals

4. System Administrator	G9: Ensure system availability and performance G10: Maintain data security during storage and transmission G11: Support Multilingual system capabilities, specifically French & English G12: Regular system testing and updates
5. Non-Technical Users (Business Analysts, Product Owners)	G13: Interact with virtual stakeholders in preferred language G14: Access and understand generated artifacts easily G15: Provide domain specific feedback for stakeholder training G16: Save and Revisit stakeholder conversations
Total #: 5	Total #: 16

1.7 Overview:

The SRS contains a description of the VSS (Sections 2), interface prototypes (Section 3) and the specific functional & non-functional requirements pertaining to it (Sections 4,5). In addition, it contains traceability connections from the requirements to the user stories and goals (Section 6).

2. General description:

2.1 Product Overview:

Our product leverages a Large Language Model (LLM) that will act as a Virtual Stakeholder System, simulating a real-life client capable of providing live, interactive feedback. This system is intended for clients who face challenges in identifying and gathering stakeholder requirements due to the availability constraints of stakeholders. This tool will simulate the stakeholders from various profiles and provide comprehensive feedback to user queries in order to streamline the project management and requirements-gathering processes.

2.2 System Objectives & Functions:

The primary objective of the VSS is to simulate real-life communication between developers and project stakeholders. It aims to:

- Simulate virtual stakeholder interactions
- Gather, refine and validate project requirements
- Support management of projects
- Create artifacts for users, such as user stories and personas

2.3 User Classes and Characteristics:

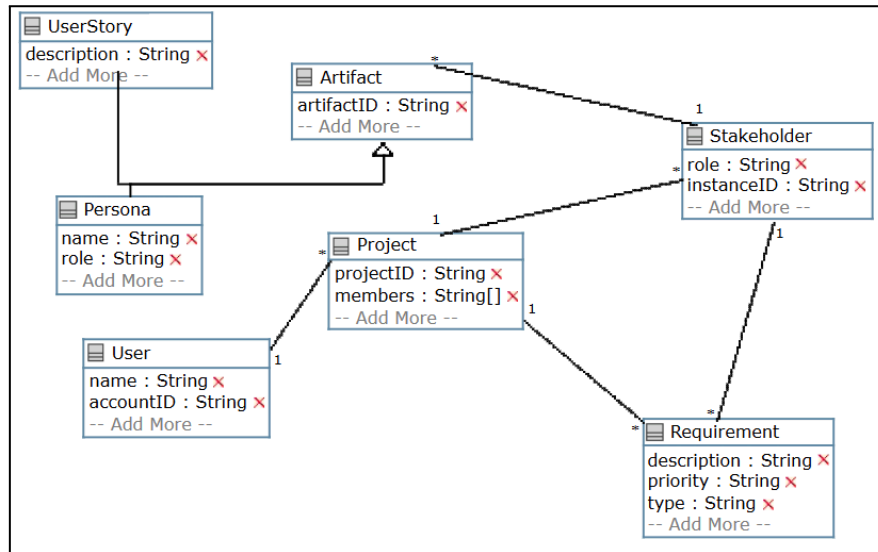


Figure 2.1: UML Class diagram

2.4 User Stories:

(US1) User Story 1: Project Manager	As a project manager, I want to make sure the needs of various stakeholders are elicited and met, in order to adjust project planning accordingly.
(US2) User Story 2: Project Team Member	As a Project team member, I want to ask the VSS for feedback on specific tasks so I can better prioritize my work.
(US3) User Story 3: Student	As a university student, I would like to use this system to assist me with writing valid requirements so that I can better learn the basics of software engineering.
(US4) User Story 4: Quality Assurance	As a quality assurance team member, I want to review current stakeholder feedback that has already been gathered, and verify that the given information is useful in order to improve the reliability of the system being built.
(US5) User Story 5: Cyber Security Specialist	As a Cyber Security Specialist, the VSS system should not compromise on any personal user data to keep the stakeholder's identities safe.
(US6) User Story 6: Hacker	As a hacker, I want to exploit unencrypted data in the system so that I can gain unauthorized access to stakeholder personal info.
(US7) User Story 7: Professor	As a Professor, I want a way to generate different personas to use as examples in my class and in exams, to help lessen my workload when building curriculums.
(US8) User Story 8: Technical	As a Technical Advisor, I want the system to provide validated

Advisor	responses and updated information in the technical domain.
(US9) User Story 9: Team Lead Coordinator	As a team lead coordinator, I want the system to produce requirement deadlines based on the project outline input by the user.
(US10) User Story 10: Software Developer	As a software developer, I want the system to produce use case scenarios that can help determine the end goals for the system.

2.5 Assumptions and Dependencies:

No assumptions or dependencies are considered at this time.

2.6 Operating Environment:

The server-side components of the software system must operate with a Linux based operating system environment.

The client side components of the VSS must operate within common web browser environments while ensuring the data at rest and in transit is encrypted by integrating firewalls and secure access protocols(SSH). The minimum set of browsers that must be supported is:

- Google Chrome -Version 90 or later
- Mozilla Firefox -Version 88 or later
- Microsoft Edge- Version 90 or later
- Safari- Version 14 or later
- Opera - Version 75 or later

3. Interface Requirements:

3.1 Interface Prototypes:

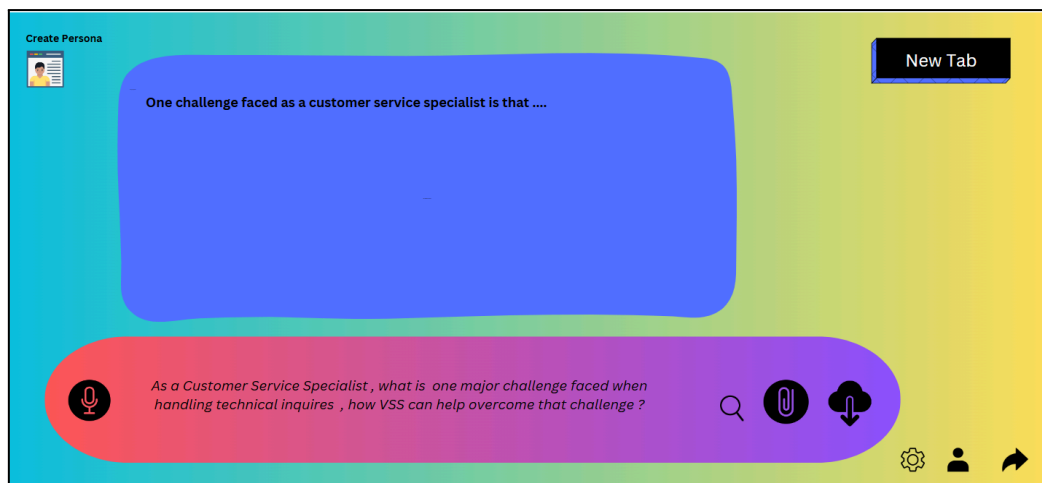


Figure 3.1: Chat bot interaction prototype

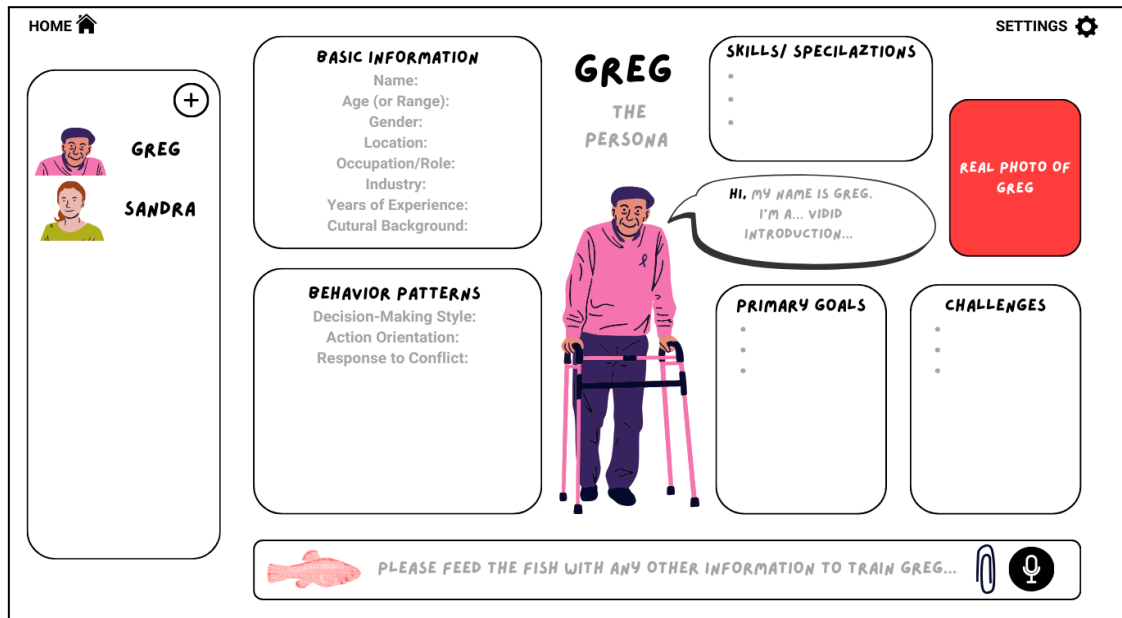


Figure 3.2: Persona Interaction Prototype

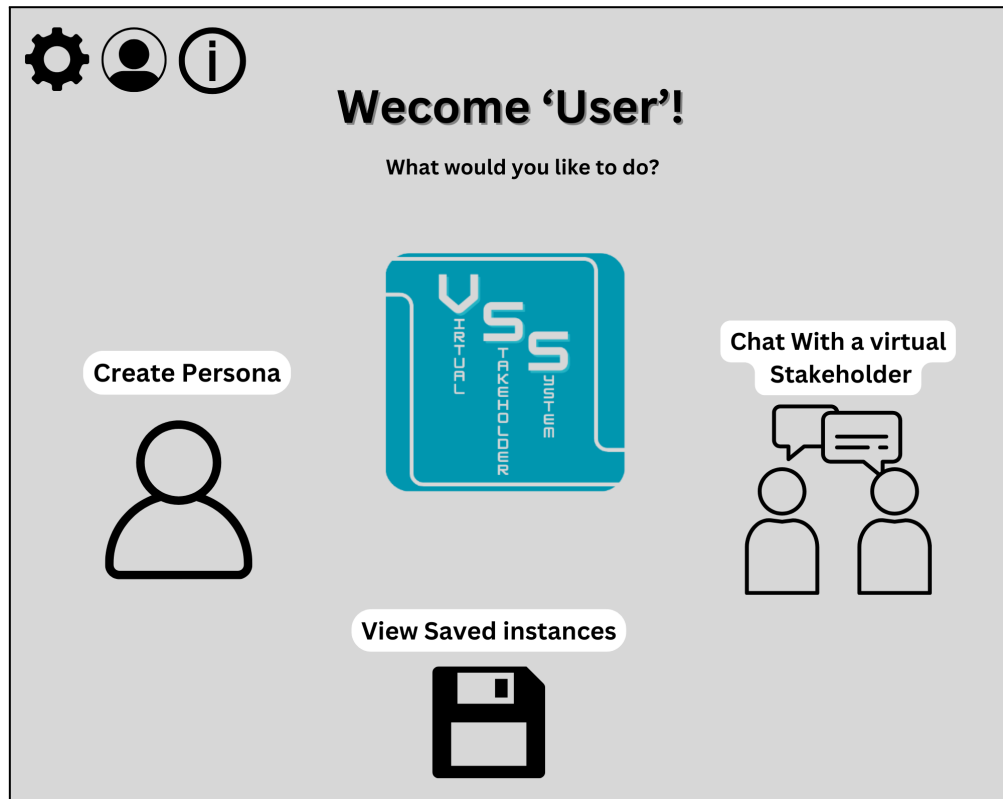


Figure 3.3: User Interface Prototype

4 . System Features:

4.1 Functional Requirements:

Requirement ID	Tag (Helix ALM)	Requirement	Guidelines (If N/A, No Guidelines Needed)	Verification Method	Stakeholder Priority
1	FR-21	The system will allow users to specify goals for their project.	Upon user request.	Demonstration	Must
2	FR-22	The system will automatically save stakeholder instances between sessions	N/A	Test	Must
5	FR-25	The system will allow users to ask the virtual stakeholder questions	By training a Large Language Model	Test	Must
25	FR-56	The system will receive the user's questions.	N/A	Test	Must
6	FR-26	The system shall provide answers to user's questions.	Allows system to act as the virtual stakeholder	Test	Must
7	FR-27	The system will allow users to create new stakeholder personas.	By training a Large Language Model	Test	Must
8	FR-28	The system will allow users to specify characteristics of created personas.	Upon user request	Demonstration	Should
9	FR-29	The system will automatically produce persona artifacts.	Upon user request	Test	Must
10	FR-30	The system will	Through data	Test	Must

		automatically update persona artifacts throughout the development process.	gathered by the LLM		
11	FR-31	The system will automatically produce user stories.	Upon user request	Test	Must
12	FR-32	The system will automatically produce non-functional requirements.	Upon user request	Test	Must
13	FR-33	The system will automatically produce use case scenarios.	Upon user request	Test	Must
14	FR-34	The system should produce requirement deadlines based on the project outline input by the user.	Deadlines should include: time (24 hours), day, month and year	Analysis	Should
15	FR-35	The system must validate certain requirements as specified by the user.	Upon user request	Test	Must
16	FR-36	The system shall automatically flag highly similar requirements.	By training a large language model on textual analysis and contextual data	Test	Must
17	FR-37	The system shall automatically flag requirements which are not coherent with each other.	By training a large language model on textual analysis and contextual data	Test	Must
18	FR-38	The system will	By allowing	Demonstration	Should

		incorporate human feedback for validation of individual or collective requirements.	users to leave feedback on requirements		
26	FR-61	The system shall maintain a glossary of project-specific terms and their definitions.	N/A	Inspection	Should
30	FR-49	The system's requirements and their corresponding deadlines shall be available to all users via a dashboard.	N/A	Demonstration	Must
31	FR-50	The system shall maintain an audit log of all requirement changes.	Should include at minimum: Who made the change, when it was changed	Test	Must
32	FR-51	The system shall allow users to export generated artifacts in common file formats.	The file export options at minimum should be: PDF, DOCX and JSON	Test	Must
34	FR-53	The system shall support tagging of artifacts.	N/A	Demonstration	Should
35	FR-54	The system shall allow users to conduct filtered searches based on tags.	N/A	Test	Should
36	FR-55	The system shall provide version history for each different iteration of a generated artifact.	N/A	Test	Should
37	FR-57	The system shall	N/A	Demonstration	Should

		provide training materials for new users.			
38	FR-58	The system shall provide technical support to users.	Through a chatbot that provides responses to user's questions	Test	Must
39	FR-59	The system shall allow users to create custom templates for generated artifacts.	Templates should support custom formatting, sections and metadata fields	Demonstration	Should
40	FR-60	The system shall provide integration capabilities with common project management tools.	Must support integration with JIRA, Trello and Azure DevOps at minimum.	Test	Must

5. Non-Functional Requirements:

5.1 Non-Functional Requirements:

Requirement ID	Tag (Helix ID)	Requirement	Guidelines	Verification Method	Stakeholder Priority
3	NFR-23	The system will allow users to save up to 15 stakeholder instances.	N/A	Test	Should
4	NFR-23	The system will fully support bilingual functionality.	English and French languages	Test	Must
19	NFR-39	The system shall ensure all information on requirements and artifacts are encrypted during storage and transmission of data.	Using industry standard security protocols	Test	Must
20	NFR-40	The system's large language model will not be trained on any personal user data.	N/A	Inspection	May
21	NFR-41	The system's virtual stakeholder must generate a response to a user's question in no more than 15 seconds.	On user request	Test	Must
22	NFR-42	The system shall produce requested artifacts in no more than 15 seconds.	On user request	Test	Must
23	NFR - 43	The system shall have 99.9% uptime	N/A	Analysis	Must

24	NFR - 44	The system will maintain an error rate of no more than 0.1% for user requests.	N/A	Analysis	Must
27	NFR - 45	The system will be tested regularly by the development team to mitigate data changes and update responses based on newly identified benchmarks	N/A	Inspection	Should
28	NFR - 47	The system's LLM will be trained with feedback from culturally diverse and gender-inclusive human reviewers to provide a balanced response aligning with the cultural values and ethics.	N/A	Inspection	Must
29	NFR - 48	The system will be trained with data from at least 50 specialized unfamiliar domains to generate responses as specialized stakeholders.	N/A	Inspection	Must
33	NFR - 52	The system shall support usage by at least 100 users without performance degradation (response time remaining under 15 seconds).	N/A	Test	Must

6. Traceability:

6.1 Traceability of Requirements to Goals & User Stories:

Requirement ID	Requirement Description	Linked Goal ID	Linked User Story ID
1	Project Goals	G1 G8	
2	Instance Saving	G16	
3	Instance Saving Limit	G16	
4	System Language	G11 G13	
5	Virtual Stakeholder Questions	G1 G7 G13	US 2
6	Virtual Stakeholder Answers	G1 G7 G13	US2
7	Persona Creation	G2	US7
8	Persona Specification	G2	US7
9	Persona Artifacts	G2 G14	US7
10	Persona Updating	G2	US7
11	User Stories	G2 G14	
12	Non Functional Requirements	G6	US3
13	Use Case Scenarios	G2 G14	US10
14	Project Deadlines	G4	US9
15	Requirement Validation	G1 G8	US4 US1
16	Duplicate Requirements	G3	US1
17	Requirement Coherency	G3	US1
18	Requirement Feedback	G15	US1 US8

19	System Data Encryption	G5 G10	US5 US6
20	LLM Data Encryption	G5 G10	US5 US6
21	Virtual Stakeholder Response Time	G7 G9	US2
22	Artifact Production Time	G9	US4
23	System Uptime	G9	US5
24	System Error Rate	G3	US5
25	Receive questions	G7 G13	US2
26	Glossary	G14	US3
27	System Testing	G12	
28	LLM Bias	G12 G15	
29	LLM Domains	G12	
30	Requirement Deadline Dashboard	G4	
31	Requirement Change Log	G5 G10	
32	Exporting Artifacts	G2	
33	Performance Degradation	G9	
34	Artifact Tagging	G1 G14	US4
35	Artifact Filtering	G1 G14	US4
36	Version Comparison	G2 G14	US10
37	Training Materials	G14	US3
38	Technical Support Chatbot	G9 G11	US8
39	Custom Templates	G4 G14	US1 US9
40	Project Management Integration	G9	US9 US1

Team Description and Roles:

Who They Are?	Tasks	Percent
Anna	- <i>Non functional requirements, interface requirements, user classes, purpose, abbreviations, scope, table of contents.</i>	20%
Alex	- <i>Non functional requirements, interface requirements, user classes, purpose, abbreviations, scope, table of contents.</i>	20%
Caleb	- <i>Non functional requirements, interface requirements, user classes, purpose, abbreviations, scope, table of contents.</i>	20%
Owen	- <i>Non functional requirements, interface requirements, user classes, purpose, abbreviations, scope, table of contents.</i>	20%
Mukhveer	- <i>Non functional requirements, interface requirements, user classes, purpose, abbreviations, scope, table of contents.</i>	20%